The Ephemeris

The Official Publication of the San Jose Astronomical Association

March 2025
Volume 36 Issue 1

Member Appreciation

Gravitational Lensing

Planetary Parade

San Jose Astronomical Association
SJAA 70 YEARS
EST. 1954

Top: Four-Panel Mosaic of Horsehead and Flame Nebulae by Carl Svensson. 38 Hours of exposures in January 2025 through narrowband filters. SV130, ASI2600MM Pro, AP 1200GTO from San Jose.

Front Cover: A Deeper Look at Rosette Nebula by Imran Badr. Over 19 hours of exposures from November 2022 and January 2025 through narrowband filters with RGB stars. Mix of equipment including ASI2600MC Pro and ASI2600MM Pro, Sky-Watcher Espirit 100ED, and ZWO AM5 mount.

Back Cover: The Christmas Tree Region by Francesco Meschia. Over 33 hours of exposures through narrowband filters. Astro-Tech AT130EDT, ASI294MM Pro, and AP Mach-1 GTO.

Inside This Issue

SJAA Contacts

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Binocular Stargazing	Ed Wong
Fix-it Program	Larry Zurbrick
Hands on Imaging	–
Imaging SIG	Hy Murveit
Astronomy 101	Carl Svensson
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Loaner Program	Muditha Kanchana
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Refreshments	Marianne Damon
Social Manager	Beth Johnson
Speakers	Chris Gottbrath
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@sj_astronomy @sjastronomy

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly. Articles and Observation Reports for publication should be submitted by e-mail to ephemeris@sjaa.net. Please only submit original content (articles or photos) of your own authorship. Copyrighted content from other authors, including images, will be rejected.

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
http://www.sjaa.net/contact

Calendar

Board Meetings + Speaker
Saturday 3/15, 4/5, 5/10, 6/7
Hogue Park, 6:00 - 7:30pm
Guest speaker follows meetings

Solar Sunday & Fix-It
Sunday 4/6, 5/4, 6/1
Hogue Park, 2:00 - 4:00pm

In-Town Star Party
Friday 3/15, 4/18, 5/23, 6/20
Hogue Park, time varies

Starry Nights Star Party
Saturday 3/22, 4/19, 5/17
RCDO, time varies

Imaging SIG
Tuesday 3/18, 4/15, 5/20, 6/17
Online, 7:30pm

Binocular Stargazing
Saturday 6/14
Coyote Valley, 9:30pm

Please refer to the SJ Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events) where you can find specific event times, locations, maps and possible cancellation due to weather.

Member Programs

SJAA LIBRARY

SJAA offers another wonderful resource: a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may send all your questions/ comments to **librarian@sjaa.net**.

TELESCOPE FIX-IT SESSION

Fix-It Day, sometimes called the Telescope Tune Up or the Telescope Fix-It program, is a really simple service the SJAA offers to members of the SJAA community for free, though it's priceless. Headed up by Larry Zurbrick, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

http://www.sjaa.net/programs/fixit/

QUICK START PROGRAM

The Quick STARt Program, headed up by Jonathan Lawton, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick STARt sessions are generally held quarterly. The Quick STARt program has resumed under its new lead, Jonathon Lawton.

http://www.sjaa.net/programs/quick-start/

LOANER PROGRAM

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick STARt program please fill this form to request a particular item. Please also consider donating unused equipment.

http://www.sjaa.net/programs/loaner-telescope-program/

ASTRO IMAGING SIG

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Hy Murveit, the Imaging SIG meets roughly every month over Zoom, often on the 3rd Tuesday evening of the month. We usually have a guest speaker presenting on some topic related to astro-imaging. Talks begin at 7:30pm. There is an opportunity to socialize and discuss imaging starting at 7pm when we start the Zoom call. Meeting topics, dates and times are posted on the SJAA Astro Imaging mail list in Google Groups. Past meetings are available on **YouTube**.

http://www.sjaa.net/programs/imaging-sig/

ASTRO IMAGING WORKSHOPS AND FIELD CLINICS

Not to be confused with the Imaging SIG group, this newly organized program is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.

Workshops are held at Hogue Park once per month and field clinics (members only) once per quarter at a dark sky site. Keep an eye on the Google Groups and website for information about the next session.

Public Programs

ASTRONOMY 101

Astronomy 101 covers various topics related to beginner astronomy and star gazing. The popular Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Hogue Park. This is a regular, monthly session, run by Carl Svensson. Content varies from "what's up in the night sky" to basics of star-gazing and telescopes to specific topics of interest. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

http://www.sjaa.net/programs/beginners-astronomy/

San Jose Astronomical Association
SJAA

SCHOOL STAR PARTY

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill. Contact SJAA's Program Coordinator for additional information.

http://www.sjaa.net/programs/school-star-party/

SOLAR OBSERVING

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time and location is subject to change.

http://www.meetup.com/SJ-Astronomy/

IN-TOWN STAR PARTY (ITSP)

This popular public star party is held every month, at our headquarters in Houge Park in San Jose. Enjoy celestial wonders from our very own backyard. Our volunteers are more than happy to show you the grate views through their telescopes. Check our Meetup page for specific dates and times. All are welcome!

BEGINNER ASTRONOMY FORUM

This exclusively online event is an opportunity for anyone to come and ask questions about the early stages of astronomy and star-gazing as a hobby. Unlike our other Astronomy 101 offers, this session is entirely guided by your questions, so come prepared! We will have two or three SJAA volunteers on-hand to share their knowledge.

STARRY NIGHTS

SJAA partners with the Santa Clara County Open Space Authority (OSA) to co-host monthly public star parties, called Starry Nights. These events are held at Rancho Canada del Oro (RCDO) Open Space Preserve just 30 minutes south of downtown San Jose. Come enjoy a night sky that is sheltered from city lights. Space is limited, so reserve your spot at:

http://www.meetup.com/SJ-Astronomy/

BINOCULAR STARGAZING

Want to learn the night sky? Did you know that you don't need to spend a ton of money on a telescope to do so? All you need is a decent pair of binoculars. Bring your 10x50mm binoculars, some comfy clothes, a chair, and join us on a tour of the night sky!

http://www.sjaa.net/events/binocular-stargazing/

San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243
(Must be 18 years or older to apply)

New Membership or Renewal?

(check one)

☐ **NEW**

☐ **RENEWAL**

Membership Type

(check one)

☐ **$20 Membership with online Ephemeris**

☐ **$30 Membership with printed Ephemeris**
(mailed to address below)

NAME ______________________________

ADDRESS ______________________________

CITY/STATE/ZIP ______________________________

PHONE ______________________________

E-MAIL ______________________________

Bring this form to any SJAA meeting or mail it to: SJAA, P.O. Box 28243 San Jose, CA 95159-8243
This newsletter is available online at http://www.sjaa.net/sjaa-newsletter-ephemeris/
Questions? Send e-mail to: memberships@sjaa.net. Join or renew online at http://www.sjaa.net/join-the-sjaa/

Member Appreciation Night

SJAA is entirely run by volunteers. From the board and leadership, to the docents at star parties, and the people serving up delicious snacks and food at events, volunteers run the show. This year, SJAA held its annual Member Appreciation Night on February 15th, just one day after Valentine's Day. What a fitting date to recognize some of our volunteers. And hey, since we're all here, how about some pizza?

Following the February board meeting, festivities kicked off at 8PM. With the aroma of freshly baked dough, cheese, and a variety of accoutrements wafting through the air, hungry amateur astronomers descended on the buffet tables. The snacks team were out in full force and put on a double-threat of a meal: orderly and delicious. As many a mouth munched on morsels, President Larry Zurbrick took the mic to address the membership.

Recipients of the SJAA Outstanding Service Award this year were introduced in pairs, mostly. First up were George and Creighton Voon. This father-son team shows up to every ITSP. Creighton mans the check-in table, which is the first point of contact for many attendees. George and Creighton also donate their time to solar observing events, and have also started to get involved with the ongoing "website refresh" effort. Walter and Frances Yau were recognized next for their work on the snacks team. Food plays an integral part in our SJAA meetings. Public talks begin with a social hour, lubricated by the snacks and refreshments provided by Walter, Frances, and the snacks team. Our potlucks and pizza parties are also augmented by the snacks team. Where there's fun, there's food, and that's often thanks to Walter and Frances.

This year also saw one recipient whose recognition had been sidelined by the COVID years. Gary Mitchell was honored with an Outstanding Service Award, originally slated to be awarded in 2020. Gary is a fixture at School Star Parties, ITSPs, and Starry Nights events, often wowing the public with views through his Meade telescope. His encyclopedic knowledge of the cosmos, coupled with his enthusiasm, makes him a stand-out volunteer and an excellent reflection of SJAA's values. Gary's scale model of the Solar System is a big hit with beginners and seasoned astronomers. Explaining astronomical distances is no small feat, but Gary's tour of the Solar System through his scale model does the job in a fun and informative way.

Many thanks to all of our Outstanding Service Award recipients! Their time, and all of our volunteers' time and effort, is the life-blood of SJAA. Without them, we wouldn't be the thriving community we are today.

By the way, did you know that there are plenty of ways that you can volunteer your time with SJAA? In addition to feeding the membership and operating a telescope at star parties, members can volunteer their time for any number of positions and opportunities. Like writing? The Ephemeris would love to feature your articles. How about talking? Astro 101 is actively looking for presenters for educational talks. Ever look at the Sun, and live to tell the tale? Our solar program is looking for volunteers this merry month of March. Are you an expert astro imager? We are in need of a new lead for the Hands-On Imaging program.

No matter your skillset, there's a role for you. Email the volunteer chair to get started at volunteerchair@sjaa.net. Maybe you are already volunteering. If so, did you know that many employers will match your volunteer hours with donations? It's true! List Rob Jaworski (treasurer@sjaa.net) as the SJAA contact if you request a matching grant from your employer.

Thank you again to all of our volunteers and members!

George and Creighton Voon are recognized for their outstanding service. Image credit: Muditha Kanchana

Larry recognizes Walter and Frances for their outstanding, and delicious, service. Image credit: Muditha Kanchana

Nothing brings friends together like a pizza party. Image credit: Muditha Kanchana

Larry presents Gary with a long-overdue award for outstanding service. Image credit: Muditha Kanchana

Observe Jupiter Like Galileo

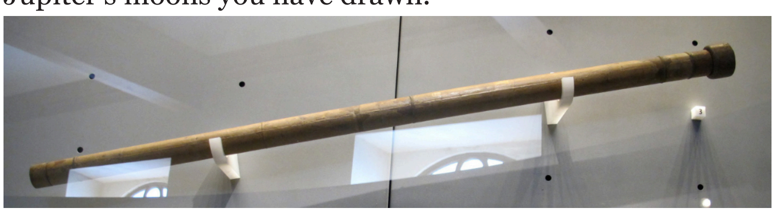
By Jordan Makower

While Jupiter is nicely positioned this quarter, use binoculars or a small telescope to see and record the movement of four of its moons; Io, Europa, Ganymede, and Callisto. This is exactly what Galileo did in 1610. Although his first telescope had only a magnifying power of 3x, he constructed one with a power of 20 to do his early observations of Jupiter.

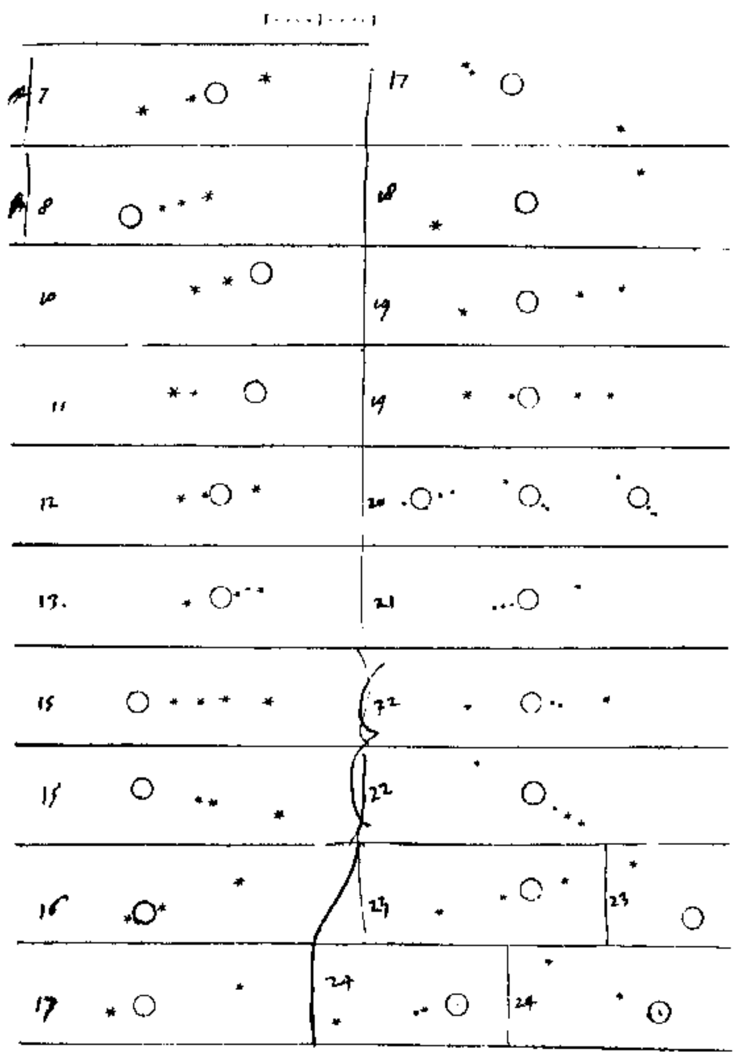
An important feature of the 20x telescope was that it had a long focal length: about 1330 mm. Even though its aperture was only about one inch, the focal length produced an image of about a quarter of a degree, which is half the apparent diameter of the Moon. With it, he was able to see detailed features on the lunar surface. He used it to view (among other things) Jupiter, Saturn, stars, and Venus. Those observations allowed him to formulate his theory that our planet and others revolved around the Sun.

Pictured at right is a copy of a page from Galileo's notebook, showing his observations in January 1610. Galileo had considerable difficulty in recognizing the true meaning of what he was seeing; Callisto often lay outside the (restricted) field of view of his telescope, Io was often lost in Jupiter's glare, and some moons occasionally disappeared in Jupiter's shadow or behind or in front of the planet itself.

Although the telescope Galileo used boasted a long tube and focal length, an average pair of modern 8x50 binoculars or a small telescope will work almost as well. Make your own sketches, indicate time and date, try to determine which of Jupiter's moons you have drawn.



Galileo's "cannocchiali" telescopes at the Museo Galileo, Florence. By Sailko - Own work, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=31258183



Galileo's drawings of Jupiter and its four biggest moons, as he observed them on various dates between January 7th and 24th, 1610. Image credit: https://www.astro.umontreal.ca

Observing Highlights

03/14 — **Total Lunar Eclipse**
A total lunar eclipse occurs when the Moon passes completely through the Earth's dark shadow, or umbra.

03/20 — **March Equinox**
Also called the spring, or vernal equinox in the Northern Hemisphere. The equator will have as much day as night.

04/21 — **Mercury at Greatest Elongation**
This is the best time to view Mercury, because it is furthest from the sun, and highest in the sky.

04/22 — **Lyrids Meteor Shower**
The Lyrids is an average shower, usually producing about 20 meteors per hour at its peak. Look for it on the 22nd and 23rd.

05/06 — **Eta Aquarids Meteor Shower**
An above average shower, capable of producing up to 30 meteors per hour. Look for it on the 6th and 7th.

05/07 — **Venus at Greatest Elongation**
The planet Venus reaches greatest eastern elongation of 45.9 degrees from the Sun. This is the best time to view Venus.

New Moon: 3/29, 4/27, 5/27, 6/11

Full Moon: 3/14, 4/13, 5/12, 6/25

Source: http://www.seasky.org/astronomy/astronomy-calendar-2025.html

Kids Corner

Word Search

- GALAXY
- MILKY WAY
- WHIRLPOOL
- PINWHEEL
- SOMBRERO
- ANDROMEDA
- NEEDLE
- CARTWHEEL
- BUTTERFLY

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Y Y V Z N X V L X K T C D K N
G A L A X Y P G W B K A P X U
K I H V M M G Y W G A R N V P
J W H I R L P O O L N T E L J
A P I N W H E E L Y D W E N L
Q F Y A I F Y A X V R H D C H
D V F J X S X E Y C O E L S W
F U F E G Z R D I H M E E O U
A K R W B Q U Q A R E L D M G
V M D A G E H E I X D D T B Z
H D S C H T K Q L P A S G R A
K M I L K Y W A Y D F Y O E P
U B N N G F B A R T Z F S R P
B M D W I L E V E O G L O O X
G B U T T E R F L Y A Q C W C
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Fun Facts

- The word 'galaxy' comes from the Greek word 'galaxias' which means 'milky'. It references our galaxy: The Milky Way.
- Our Milky Way Galaxy is about 87,000 light years across, but only 1,000 light years thick.
- The Milky Way weighs in at 1.5 trillion solar masses! A "solar mass" is equal to the mass of the Sun. That's heavy!
- The Milky Way contains at least 100 billion planets!

Sources: wikipdeia, science.nasa.gov

Elections and Appointments

Annual board member elections were held this year between February 8th and 15th. Five people were up for election for five board positions. All candidates received sufficient votes to be elected. Ken Miura, Swami Nigam, and Marianne Damon were re-elected to their board positions, with Larry Zurbrick replacing Vini Carter in the Director 1 seat. Last quarter also saw the appointment of two positions by the board. Carl Svensson was appointed to Vice President, filling the previously vacant position. Chris Gottbrath took on the Speaker Chair and Volunteer Chair roles, while Steven Green steps up to the Mendoza Ranch program lead with Doug Loyer as backup.

SJAA By The Numbers

Dec 2024 - March 2025

Membership	441	(+48)
X Followers	901	(-44)
Facebook Followers	836	(+7)
Blue Sky Followers	444	(+158)
Instagram Followers	169	(+7)

Spring Swap Meet 2025

The SJAA Swap Meet back in September was an all around success. The powers that be decided we should do it again, and we're ready to announce the date: **Sunday, 16 March 2025 .**

As before, this event will be held at SJAA basecamp Houge Park in San Jose from 9am to about noon.

The format of this swap will be very laissez faire. There won't be much/any overhead such as registration, tracking, etc. There will be a few tables set up in the hall along with some chairs, the doors will swing open and the local amateur astronomy community can use this as an opportunity to clean out their astro closets, or pick up a few things that might catch their eye.

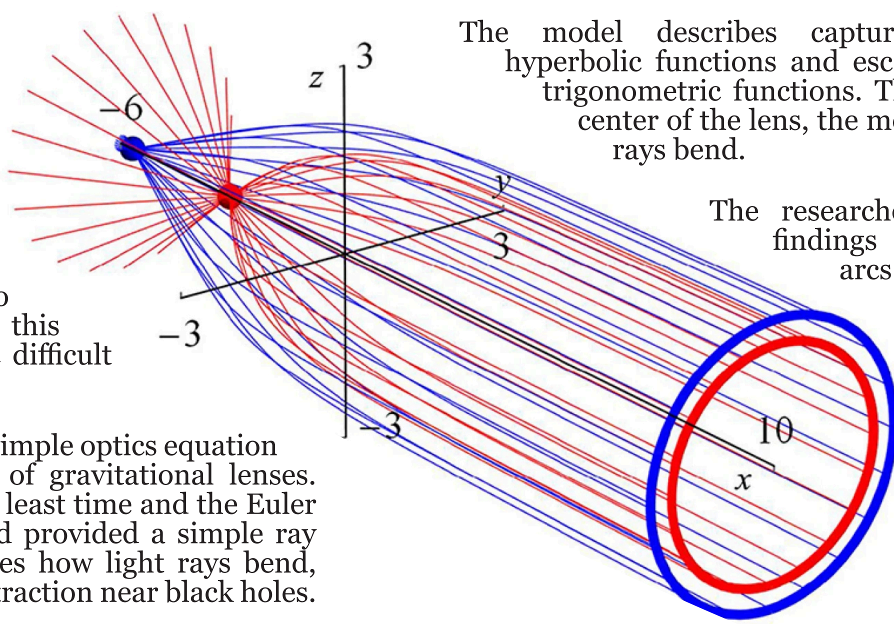
Again, doors open at 9am for sellers, buyers and browsers. More information and details can be found on the SJAA website here:

https://www.sjaa.net/events/swap-meet/

Model for Gravitational Lensing

SJAA's own Bogdan Szafraniec has had their paper published in the American Journal of Physics. The full work is entitled, "A simple model of a gravitational lens from geometric optics," and can be found at https://doi.org/10.1063/5.0157513. The following summary is from a press release via aip.org: https://ww2.aip.org/scilights/optics-equation-provides-approximation-of-gravitational-lenses-for-students

Gravitational lensing is the phenomenon of light bending around massive objects along curves in space-time, a major conclusion of the general theory of relativity. Typically a thorough working knowledge of Einstein's theory is required to develop an intuition about how this process works, making the topic difficult for students.



A pair of researchers proposed a simple optics equation that mimics the basic behavior of gravitational lenses. Drawing on Fermat's principle of least time and the Euler equation, Szafraniec and Harford provided a simple ray trajectory equation that illustrates how light rays bend, reflect and loop as well as fatal attraction near black holes.

"The solution described in terms of trigonometric or hyperbolic functions is so simple that with very little effort it is possible to consider very complex scenarios, such as imaging a galaxy located behind the gravitational lens," said author Bogdan Szafraniec. "It is the simplicity of the model that makes it attractive."

The model describes captured rays with hyperbolic functions and escaping rays with trigonometric functions. The closer to the center of the lens, the more the escaping rays bend.

The researchers used these findings to reproduce arcs and halo-like Einstein rings that have been observed in images from the Hubble and James Webb space telescopes.

The new model adds to other attempts to make general relativity more accessible, such as using the base of a wine glass to mimic the effects of a gravitational lens.

The authors hope their model will inspire more students to look for simple solutions as well as study general relativity.

The View from Below the Skies

WHERE ARE THE ATMS? Not the ATMs that allow access to your bank accounts but Amateur Telescope Makers. Grinding mirrors and kludging together materials on hand to build a telescope has all but disappeared with the advent of inexpensive optics and relatively low-cost, entry-level telescopes. Pretty much are gone of the days of "Amateur Telescope Making" as epitomized in the books from Albert G. Ingalls and the Scientific America Amateur Scientist articles over the decades. Ditto for building your own CCD camera from scratch using electronic components. (Does anyone remember the book "CCD Astronomy" on the topic of construction and use of an astronomical CCD camera by Christian Bull circa 1991?) Most, if not all, of the hardware that ATMs made in the past is now commercially available at a cost and quality level that makes it difficult to justify building something from "scratch."

However, amateur telescope making is not something that has past, it has evolved. The advent of readily available 3D printers, free open-source software, single board computers, inexpensive printed circuit board and machining vendors, on-line suppliers of telescope components, and maker-related web sites has changed our hobby. Professional level accessories can be 3D printed at home for the cost of a few cents to a few dollars of plastic filament. I recently came across two sets of unique binocular telescopes 3D printed by our Vice President at one of our events. One of our past [Imaging Special Interest Group (SIG) presentations]() was dedicated to 3D printing of telescope accessories. Kits such as Onstep telescope controller are available to automate telescope mounts. Free software such as Astroberry for Raspberry Pi's is available to automate your telescope and astrophotography. We are in a new golden age of amateur telescope making.

I have always been somewhat of a hands-on hobbyist and tinkerer and is evidenced by my interest in our Fix It program. Do you have an interest in getting together to discuss your projects? Let me know if you are interested in participating in an equipment night at Houge Park to show your projects or discuss what you are working on. Send me an email at fixit@sjaa.net.

The ATM is dead, long live the ATM. Ad astra.

Larry Zurbrick
President, SJAA

Board Minutes

DECEMBER 14, 2024

2025 CALENDAR APPROVED The 2025 draft calendar was emailed by Kanch and Joe. Reminders sent by Swami and Larry for SJAA personnel to review the draft calendar and provide feedback. The board approved the 2025 draft calendar.

JANUARY 11, 2025

ADJUSTING MEMBERSHIP FEE Wolf discussed the idea of possibly raising the dues to perhaps $25 later this year. However, there was quite a bit of discussion and dissent. Rob showed the club's financial statements from the past several years. We'll revisit this topic in coming meetings, after a leadership summit to plan and prioritize club goals for the next few years.

FEBRUARY 15, 2025

PARKS DEPARTMENT SJAA's ongoing relationship with the San Jose Parks Department requires ongoing reporting. The city was looking for some additional insurance related info, so Rob provided them the info they wanted. March will be the completion of the next 6 month period of the contract with the city, so SJAA will need to provide them the 6 month report. Marianne, Renuka and Rob are working on getting the report ready.

24" DOB TASK FORCE Lack of any response from OSA is very frustrating. It might be worth someone from SJAA to start attending their board meetings, which can be joined online. Larry is willing to attend their board meeting.

SANTA CRUZ STEAM EXPO Wolf is leading a group of SJAA volunteers to provide solar observing services at the Santa Cruz County STEAM Expo in March. We currently have only seven fully confirmed volunteers. We need a few more volunteers. Some tentative folks may still come through, but if anyone else wants to sign up for sure, please speak up! It should be a good event.

Collaboration with Astro Revolution Work continues with the students from Astro Revolution. They are interested in holding a School Star Party for Greystone Elementary, and are poised to give an intro talk at one of our Astro 101 events before ITSP. Additional volunteers identified for ITSP. *Ed. note: Astro Revolution gave their talk to an appreciative audience on February 21st.*

DISCORD FOR SJAA Carl proposed using the popular community chat application, Discord for SJAA. The server is provided to the Board for evaluation, and Carl will come up with a plan to roll out to the general membership.

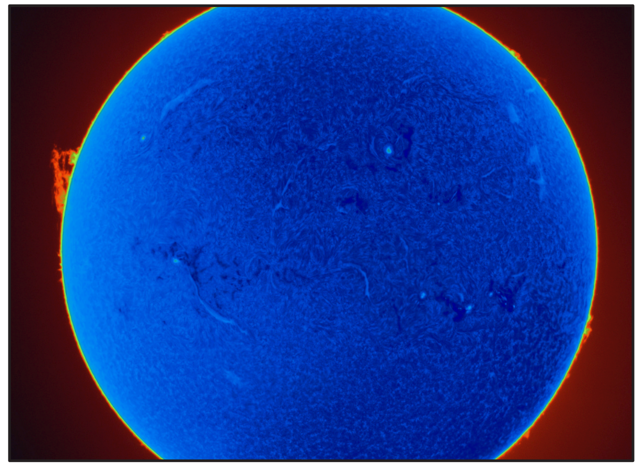
Excerpts from this quarter's board meetings are above. Full minutes are available online at https://www.sjaa.net/about/board-meeting-minutes/

Planetary Parade

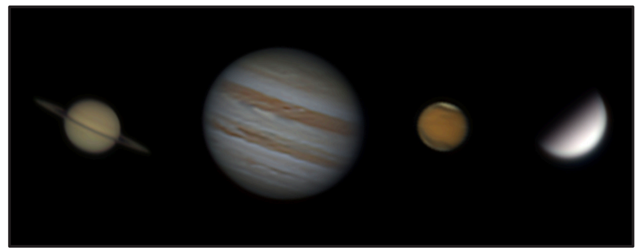
January and February saw a "grand alignment" of all the planets in the Solar System. Mercury, Venus, Mars, Jupiter, Saturn, Uranus, and Neptune (sorry, Pluto) all became visible at the same time in the night sky. Early in the evenings, keen-eyed observers could spot Mercury and Saturn low on the horizon, trailing the setting sun, followed by Neptune, Venus, Uranus, Jupiter, and Mars.

All planets orbit the Sun in roughly the same plane. This is due to the fact that all the planets formed from the same disc of gas and dust that surrounded the infant Sun. This proto-planetary disc defined the path that all planets orbit the Sun, which we call the Ecliptic. This is why you can often see planets "align" in the night sky: they are already mostly aligned, and always have been.

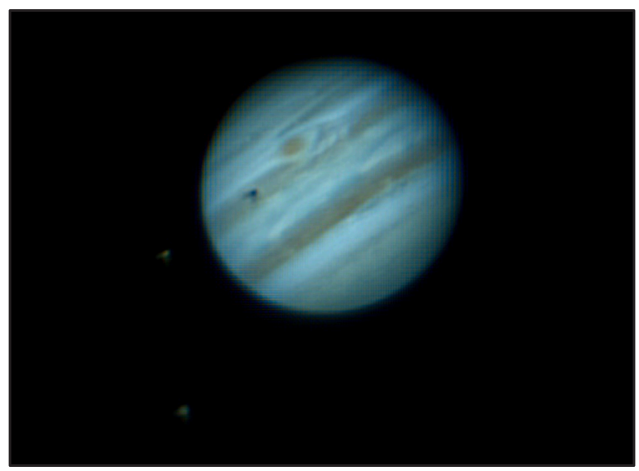
Members of SJAA diligently photograph Solar System objects, and their efforts are showcased on this page. Jupiter often presents the sharpest images with the most detail, due to its sheer size. Saturn and Mars are not far behind, under the right conditions. Even Venus, though mostly covered in opaque clouds, presents an interesting target as it goes through phases, just like the Moon.



This false-color image of the Sun by Mark Scrivener highlights edge-on prominences, or filaments, and the solar surface in blue, with a dramatic ring of fire in red. The profile of a prominence can be seen erupting from the left. Taken with a Lunt60THA filter.

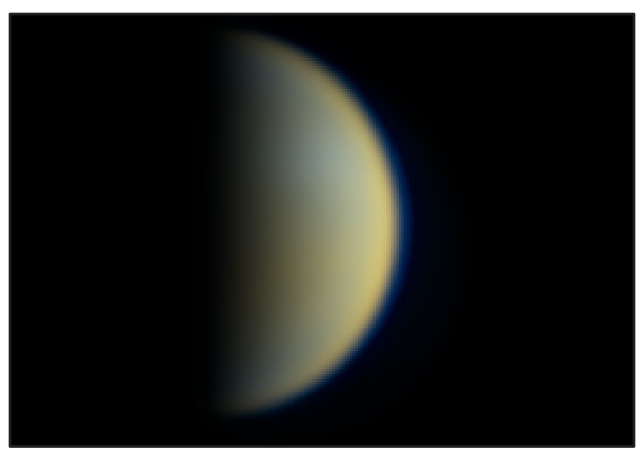


Chih-hsiang Cheng (Shawn) rang in the New Year by capturing the big four planetary image targets using an AT115EDT and ASI183MM Pro.



Left: Francesco captured Jupiter on December 8th, 2024 with an AT130EDT refractor, Televue Powermate 2x Barlow, ASI294MM Pro and RGB filters. This is a stack of 3750 frames of 15ms each.

Right: Carl imaged Jupiter along with it's Great Red Spot, Io's shadow, and Io and Ganymede themselves.



Left: Rich Klein captured this image of Mars made up of 600 subframes taken through an 8" SCT telescope.

Right: This image of Venus by Christos Gougoussis shows it in one of its crescent phases. Pale variations in cloud cover can be seen as different shades of yellow and gray.